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# Advanced Programming

ÁRPÁD TAMÁSI · ELTE FACULTY OF SOCIAL SCIENCES

SESSION 2 — 15 SEPTEMBER 2026

# How the grade is made

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Homework\*

40% — due Friday 20:00

Final demo

40% — at the last two sessions, **your classmates are the jury**

Instructor

20% — my own assessment

**\*Late:** by Monday 20:00 it counts **half** · after that, **not at all**

*The full rules: the **Requirements** page on Canvas.*

# Today's session

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|   | MIN | WHAT                                              |
|---|-----|---------------------------------------------------|
| A | 40  | Your environment: VS Code, an agent, your project |
| B | 30  | Vibe coding and agentic engineering               |
| C | 15  | The homework                                      |

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# Your environment

# Form groups

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- **Each row is a group — help each other**
- **Stuck? Raise your hand**

# What you are putting together

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| WHAT           | WHAT IT DOES                                                  |
|----------------|---------------------------------------------------------------|
| VS Code        | Where you work — files, editor, terminal                      |
| A coding agent | Claude Code or Codex, inside VS Code — it works on your files |
| Your project   | Your repository, on your own machine                          |

**The goal for today: your project is open, and the agent can tell you what it is.**



# 1. VS Code

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1. **code.visualstudio.com** → the big download button
2. **File → Open Folder** — always open a folder, never a single file
3. **Trust** the folder — in *Restricted Mode* the agent cannot work
4. **Cmd+Shift+P · Ctrl+Shift+P** — the command palette

## 2. An agent inside VS Code

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**Cmd+Shift+X · Ctrl+Shift+X → search → Install → Sign in.** One is enough. Then open its own panel — not VS Code's built-in Chat.



### Claude Code for VS Code

Claude Code for VS Code: Harness the power of Claude Code without leaving your IDE

 Anthropic

NEEDS A PAID CLAUDE PLAN — PRO OR ABOVE



### Codex – OpenAI's coding agent

Codex is a coding agent that works with you everywhere you code — included in ChatGPT Plus

 OpenAI

NEEDS A CHATGPT ACCOUNT — THE FREE TIER IS LIMITED



### 3. Let the agent install the rest

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An empty folder, **course**, in Documents → **File** → **Open Folder**

Then, to the agent:

```
Install git and Python if they are missing.  
Where possible, without an administrator password.  
Tell me which steps I have to do by hand.
```

**What is left for you to do by hand, it tells you.**

## 4. Your project, onto your machine

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To the agent:

```
Clone https://github.com/esst-prog2/<your-repo> into this folder.
```

Then **File** → **Open Folder** → your project's folder.

*Your repository's name is on the Canvas page **Homework 1 — the projects**.*

# If git will not install

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1. Your repository on GitHub → green **Code** button → **Download ZIP**
2. Unzip it
3. **File** → **Open Folder** → that folder

Today's goal is still met. Git can wait for next week.

## 5. Look around

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With your project open, to the agent:

Look around

It tells you about your project, from your README. **That is the check.**

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# Vibe coding and agentic engineering

# “Vibe coding”

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Andrej Karpathy · February 2025 · the post that named it

*There's a new kind of coding I call “vibe coding”, where you fully give in to the vibes, embrace exponentials, and forget that the code even exists.*

*I “Accept All” always, I don't read the diffs anymore.*

*It's not too bad for throwaway weekend projects.*



# The same thing, named precisely

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Simon Willison · October 2025

*Vibe coding: the fast, loose and irresponsible way of building software with AI — entirely prompt-driven, and with no attention paid to how the code actually works.*

He proposed a name for the other thing: **vibe engineering** — professionals who speed up with LLMs “*while staying proudly and confidently accountable for the software they produce.*”

# Agentic engineering

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Andrej Karpathy · Sequoia AI Ascent · 30 April 2026

*Vibe coding raises the floor. Agentic engineering is about extrapolating the ceiling.*

*Agentic engineering is about preserving the quality bar of professional software.  
You are still responsible for your software, just as before.*



# What the difference is

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## VIBE CODING

- You look at **what it does**
- The prompt is the whole plan
- An error message goes straight back in
- Done when it seems to work

## AGENTIC ENGINEERING

- You also look at **how it does it**
- A spec, then a plan, then small steps
- A failing test says what is wrong
- Done when **you can show it works**

# This course is about the second one

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**Agentic engineering.**

**Everyone takes away as much as they want.**

# A hint for using AI

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The biggest problem is not that it hallucinates. It is that what it gives you is **irrelevant**.

## RECOMMENDED METHOD

1. I think about the problem
2. I ask the AI for a **second opinion**
3. I decide — and move on

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# The homework

# This homework is vibe coding

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- **Prompts only** — do not open the code, do not edit it
- Something wrong? **Tell the agent**, in words
- **Try what you built** — play it

**Next week we look at what came out.**

# 1. Snake

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- Fork **esst-prog2/snake**
- Build it with the agent
- **Play it.** Write down what is wrong — and tell the agent



## 2. Kaszinó

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- Fork **esst-prog2/casino**
- Set it up, run the tests — **all red**
- Get them green with the agent
- Play a whole deal **against the computer, on a table in the browser**
- Write down what is wrong — and tell the agent

# By Friday

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1. **Snake** — your fork: the game, and a short note on what playing found
2. **Kaszinó** — your fork: tests green, the table in the browser, and a short note on what playing found

**Deadline: Friday 18 September, 20:00**

**Do not edit the tests you were given.**



# Exit ticket

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1. Is your project open, and did the agent describe it? **yes / no / partly**
2. Where did your setup get stuck, if anywhere?
3. One sentence: what is the difference between vibe coding and agentic engineering?

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Thank you